



SEMESTER END EXAMINATIONS – JULY / AUGUST 2024

Program	: B.E. – Common to CSE / ISE / CSE(CY) / AI & DS / BT/ AI & ML/ CSE (AI & ML)	Semester	: II
Course Name	: Numerical Techniques and Differential Equations	Max. Marks	: 100
Course Code	: MAC21	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define algebraic & transcendental equations with an example. CO1 (02)
 - Expand $\tan^{-1}x$ in powers of $(x-1)$ up to the term containing fourth degree term. CO1 (04)
 - Find the extreme values of $f(x, y) = \sin x \sin y \sin(x+y)$; $0 < x < \frac{\pi}{2}, 0 < y < \frac{\pi}{2}$. CO1 (07)
 - Solve the system of non-linear equations $x^2 + y^2 = x$, $x^2 - y^2 = y$, given that $x_0 = 0.8$ and $y_0 = 0.3$ using Newton-Raphson method (Carry out two iterations). CO1 (07)
- Write Maclaurin's series for the function of one variable. CO1 (02)
 - Find a negative root of the equation $xe^x - \sin x = 0$ using Newton-Raphson method correct to three decimals. CO1 (04)
 - Expand the function $e^y \log x$ about $(1, -3)$ up to second degree terms. CO1 (07)
 - Find the maximum and minimum distances of the point $(2, 3, 8)$ from the sphere $x^2 + y^2 + z^2 = 56$. CO1 (07)

UNIT - II

- Write Euler's formula to solve first order and first degree ODE. CO2 (02)
 - Using Taylor's series method solve the initial value problem at $t = 0.1$ given that $\frac{dy}{dt} = e^{-2t} - 2y$; $y(0) = \frac{1}{10}$ considering derivatives up to 3rd degree. CO2 (04)
 - Using Runge-Kutta method of fourth order, find y at $x = 0.2$ taking $h = 0.2$, if $\frac{dy}{dx} = x^2 + y^2$; $y(0) = 1$. CO2 (07)
 - Obtain the orthogonal trajectories of the family of curves $x^2 + y^2 - 2a^2 \log x = c$ where c is the parameter. CO2 (07)
- Write the DE of the closed-circuit involving R and C along with a voltage source E. CO2 (02)
 - If $r^n = k \cos(n\theta)$ is the orthogonal trajectory of a given family of curves, where k is a parameter, then find the differential equation of that family. CO2 (04)

- c) Use Euler's modified method to solve $\frac{dy}{dx} = \frac{-y}{1+x}$; $y(3) = 2$ at $x = 3.4$ CO2 (07)
taking $h = 0.2$ (perform two iterations at each stage).
- d) The temperature of cup of coffee is 90°C , when freshly poured and the room temperature being 24°C . In one minute it has cooled to 75°C . How long a period must elapse before the temperature of coffee becomes 60°C ? CO2 (07)

UNIT - III

5. a) Define non-homogeneous linear differential equation of 2nd order with an example. CO3 (02)
- b) If $D = \frac{d}{dx}$ and $X = X(x)$, then prove that $\frac{1}{D+a} X = e^{-ax} \int X e^{ax} dx$. CO3 (04)
- c) Solve $y'' + 2y' + 2y = e^{-x} \sec^3 x$ by the method of variation of parameters. CO3 (07)
- d) Solve $x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = x \log x$. CO3 (07)
6. a) Write the complementary function if the roots of the A.E. are $-3 \pm i, 2, 2$. CO3 (02)
- b) Solve the boundary value problem $y'' + 4y' + 4y = 0$, $y(1) = 0, y'(0) = -1$. CO3 (04)
- c) Solve $(D^3 - 3D^2 + 3D - 1)y = 2^x + x$. CO3 (07)
- d) Solve $y'' - 3y' + 5y = e^{-x} \sin^2 x$. CO3 (07)

UNIT - IV

7. a) Write Simpson's $1/3^{\text{rd}}$ rule for numerical integration. CO4 (02)
- b) From the table given below, estimate the values of $\cosh(0.56)$. CO4 (04)

x	0.5	0.6	0.7	0.8
$\cosh(x)$	1.1276	1.1855	1.2552	1.3374

- c) Use Lagrange's interpolation formula to find the value of $f(4)$ from the following data: CO4 (07)

x	-1	0	2	5
$f(x)$	9	5	3	15

- d) Find $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$ at $x = 7$ from the following data: CO4 (07)

x	1	3	5	7
y	9	11	13	63

8. a) Write the formula to find $y'(x)$ and $y''(x)$ by using Newton-Gregory forward interpolation formula when $x = x_0$. CO4 (02)

- b) Find the missing terms in the following data: CO4 (04)

x	-15	-10	-5	0	5	10
y	2	-	10	22	-	77

- c) Using Simpson's $3/8^{\text{th}}$ rule, obtain an approximate value of the integral CO4 (07)

$$\int_0^{0.3} (1 - 8x^3)^{1/2} dx \text{ by considering seven equidistant ordinates.}$$

- d) Find the cubic polynomial $y = a + bx + cx^2 + dx^3$ which passes through the points $(2, 4)$, $(4, 56)$, $(9, 711)$, $(10, 980)$ using Newton's divided difference method and hence estimate the value of y when $x = 1.5$. CO4 (07)

UNIT - V

9. a) Define rank of a given matrix. CO5 (02)
 b) Solve the system of equations $5x + 2y + z = 12$; $x + 4y + 2z = 15$; $x + 2y + 5z = 20$ using Gauss-Seidel method by taking $(1, 0, 0)$ as initial approximation. Perform two iterations. CO5 (04)
 c) The following table gives the amount in milligram of vitamin A, vitamin C and calcium contained in 1 gram of three different foods. Suppose that a dietician wants to prepare a meal that provides 200 mg of vitamin A, 250 mg of vitamin C, and 300 mg of calcium. How much of each food should be used? (Use Gauss elimination method). CO5 (07)

	Food 1	Food 2	Food 3
Vitamin A	10	30	20
Vitamin C	50	30	25
Calcium	60	20	40

- d) Solve the system of first order and first degree linear ordinary differential equations $x'_1 = x_1 + x_2$, $x'_2 = 3x_1 + 3x_2$; $x_1(0) = 0$, $x_2(0) = 1$ using matrix method. CO5 (07)
10. a) Write the conditions for consistency and inconsistency of the system of linear equations $AX = B$. CO5 (02)
 b) Prove that if λ is an Eigenvalue of a matrix A then $1/\lambda$ is an Eigenvalue of the inverse matrix A^{-1} . CO5 (04)
 c) Find the Eigenvector corresponding to the largest Eigenvalue of the matrix $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$ by Rayleigh's Power method by taking $[1 \ 1 \ 0]^T$ as the initial approximation. (Carryout five iterations). CO5 (07)
 d) Diagonalize the matrix $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$ and hence find A^8 . CO5 (07)
